Supplementary material for "HarvestMem: event-centric memory for vision–language–action harvesting robots under foliage occlusion"

This document contains the material referenced from the main text as Sections S1–S7, Tables S1–S9 and Fig. S1. Section, equation and table numbers without the prefix "S" refer to the main text.

S1. Nomenclature

B	memory budget (number of past frames kept), here $B = 16$
$c_U(t)$	coverage of a completed pick by uniform memory at decision time t
S_e, W	key set of event e (frames within $\pm W$ of its contact and release keyframes), $W = 3$
d_t, $\hat{d}_t$	ground-truth and perceived distance from the TCP to the nearest pickable fruit at frame t (m)
$\mathcal{P}_t$	set of pickable fruits still attached at frame t
k	maximum number of event keyframes reserved in the memory
r, tol	event radius and contact tolerance (m)
t^{c}, t^{r}	contact and release keyframes of an event
$\mathcal{M}_t$	set of past frames in the memory at decision time t
α, τ_d	mixing weight and distance scale of PEKM-S
σ, τ	stationary standard deviation (rad) and correlation time (steps) of the DART-H noise
DART	disturbances for augmenting robot trajectories (Laskey et al., 2017)
EV	event memory; EVU, EVR, EVA: its uniform, random-replacement and anti-event controls
FDM	fruit distance module
PEKM	picking-event keyframe memory
TCP	tool centre point
VLA	vision–language–action (policy)

S2. Benchmark details

Each episode of HarvestMem-Sim samples a tree with a trunk, branches and leaf clusters and 4–6 spherical fruits (radius 2.5–2.9 cm; heights 0.44–0.64 m; distances 0.38–0.50 m from the robot base; azimuth within ±28°). The layout sampler guarantees that each fruit has a horizontal-approach inverse-kinematics solution, that no branch blocks its approach corridor, and that fruits do not occlude each other in the front camera. A fruit is kinematically attached to its stem until it has been held between both fingers for two consecutive control steps; it then becomes a dynamic rigid body. A demonstrator attempt that fails (planning failure, no detachment, missed basket, task not completed) is discarded and the seed is advanced. Training, validation and test seeds use base seeds 100000, 200000 and 300000. Table S1 quantifies occlusion and the phase structure of the demonstrations.

Table S1: Occlusion and phase structure (clean training set).

Quantity	Value
Leaf clusters per FruitOccludedRipe episode	2 (66% hide a ripe fruit)
Occluded fruits fully invisible in the front camera after the gust	100%
Frames (fruit still on tree) in which an occluded fruit is invisible	mean 72%, median 78%
Visible fraction of non-occluded fruits after the gust (median)	1.00
Ripe / unripe fruits per FruitOccludedRipe episode	2.54 / 2.15
Mean steps per phase, FruitCount (observe / reach / approach / grasp / pull / carry / release / settle)	30 / 118 / 59 / 29 / 58 / 117 / 23 / 34

S3. Event detection: full results

Table S2 gives event detection from the ground-truth distance for every radius, and Table S3 the robustness of detection and of the K4 memory to perturbations of the validation distance signal. Perturbations: white Gaussian noise; correlated Ornstein–Uhlenbeck noise; independent dropout (each frame missing with the given probability); contiguous dropout (runs averaging 10 frames, 30% of frames in total); constant bias. Detectors: plain thresholding; causal 5-frame median filter followed by thresholding (median5); hysteresis (enter at r, leave at $1.5r$, minimum run 3 frames).

S4. Memory selection: PEKM-S details and full coverage results

PEKM-S sampling. PEKM-S draws the B memory frames from the density ρ_f of Section 4.5 over the N history frames. The per-frame density is capped

Table S2: Event detection versus radius r (tol = 5 mm, ground-truth distance). Train: 200 episodes; validation: 199 episodes with 2–5 picks each. Count: fraction of validation episodes with the exact pick count.

	Training			Validation			
r (cm)	P	R	F1	P	R	F1	Count
0.5	0.982	0.603	0.747	0.968	0.628	0.762	0.30
1	1.000	1.000	1.000	1.000	1.000	1.000	1.00
2	1.000	1.000	1.000	1.000	1.000	1.000	1.00
3 (used)	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.000**	**1.00**
4	1.000	1.000	1.000	1.000	1.000	1.000	1.00
5	0.996	1.000	0.998	0.998	1.000	0.999	0.99
6	0.971	1.000	0.985	0.962	1.000	0.981	0.91
8	0.788	1.000	0.881	0.813	1.000	0.897	0.67
10	0.647	1.000	0.786	0.692	1.000	0.818	0.49

at one slot by water-filling (excess mass is redistributed proportionally over the uncapped frames), and frames are drawn by systematic sampling of the cumulative density with a fixed offset, so the selection is deterministic given $d_{0:t}$. The first frame and the 4 most recent uniform slots are always kept, as in the base model's sampler. We evaluated $\alpha \in \{0.25, 0.5, 1\}$.

Full coverage table. Table S4 extends the main-text coverage table with run coverage (at least one memory frame inside the pick's near run), $k = 12$ and PEKM-S with $\alpha = 0.5$. Figure S1 shows the frames selected by each memory in one validation episode.

S5. Perception: illumination and anchors

Prompt design. The SAM prompt (bounding box padded by 2 pixels plus the component centroid as a positive point) was selected on rendered training episodes, where it reached an instance F1 of 0.991 and a mean IoU of 0.970. HSV thresholds were calibrated on 20 rendered training episodes (pixel F1 0.986 for ripe and 0.976 for unripe fruits). Fruits with fewer than 20 (front) or 60 (wrist) visible pixels are ignore regions.

Illumination shifts. Every third frame of the perception set was re-rendered photometrically as overcast or dusk ($0.55\,x^{1.2}$), direct sun ($1.35\,x^{0.85}$) and

Table S3: Robustness of event detection and K4 memory to perturbations of the validation distance signal (5 seeds per level). Each cell: plain / median5 / hysteresis. Keyframe recall: fraction of clean-signal keyframes found within ±3 frames in the perturbed memory.

Perturbation	Level	Event F1	Count exact	Keyframe recall (K4)
None	–	1.000 / 1.000 / 1.000	1.00 / 1.00 / 1.00	1.000 / 1.000 / 1.000
White noise	0.5 cm	0.971 / 1.000 / 1.000	0.86 / 1.00 / 1.00	0.985 / 0.989 / 0.997
White noise	1 cm	0.846 / 0.999 / 0.999	0.37 / 0.99 / 0.99	0.932 / 0.980 / 0.995
White noise	2 cm	0.635 / 0.969 / 0.947	0.03 / 0.85 / 0.78	0.846 / 0.966 / 0.958
Correlated (OU)	2 cm	0.887 / 0.941 / 0.953	0.58 / 0.74 / 0.79	0.918 / 0.926 / 0.931
Independent dropout	30%	1.000 / 1.000 / 0.998	1.00 / 1.00 / 0.99	0.998 / 0.999 / 0.996
Independent dropout	50%	0.996 / 0.996 / 0.988	0.98 / 0.98 / 0.94	0.982 / 0.987 / 0.973
Contiguous dropout	30%	0.913 / 0.913 / 0.894	0.62 / 0.62 / 0.55	0.882 / 0.901 / 0.867
Bias	−1 cm	1.000 / 1.000 / 1.000	1.00 / 1.00 / 1.00	0.998 / 0.999 / 0.999
Bias	+2 cm	1.000 / 1.000 / 1.000	1.00 / 1.00 / 1.00	1.000 / 1.000 / 1.000

warm low sun (channel gains 1.12 / 1.00 / 0.80), with the colour thresholds kept as calibrated under nominal lighting (Table S5).

Anchors. Table S6 reports the frame-0 anchors built by SAM-FDM, matched to the true fruit positions at frame 0 with a 5 cm gate. Depth noise follows $\sigma(z) = 1\ \text{mm} + 2.5\ \text{mm} \cdot z^2$ (z in m) with 2% missing pixels, typical of consumer RGB-D cameras.

S6. Policy analyses: open-loop accuracy and memory ablations of the base VLA

Table S7 evaluates a uniform-memory policy trained on clean demonstrations in open loop (teacher forcing at the demonstrated states). Table S8 summarizes memory ablations of the base VLA on a general manipulation-memory benchmark ([TBD: cite RoboMME]), evaluated under sealed protocols with paired bootstrap and Holm adjustment.

S7. Reproducibility: parameters, metric definitions and implementation

Parameter provenance. Every parameter was fixed before the formal evaluation, either on training data or from physical or architectural considerations (Table S9).

Table S4: Information retained by 16-frame memories (validation, both tasks; 5318 decisions, 7841 decision–pick pairs). Last / 2nd / 3rd: coverage of the most recent, second and third most recent pick. Near: fraction of memory frames with $d_t \leq r$. Gap: median gap between memory frames (frames).

Memory	Event cov.	Run cov.	Count rec.	Last	2nd	3rd	Near	Gap
Uniform (U, K0)	0.570	0.461	0.431	0.675	0.472	0.416	0.042	14
Recent	0.080	0.069	0.046	0.145	0.000	0.000	0.049	209
Random	0.432	0.364	0.329	0.533	0.351	0.231	0.040	41
PEKM $k = 2$	0.729	0.681	0.579	1.000	0.430	0.336	0.085	17
PEKM $k = 4$	**0.876**	**0.875**	**0.814**	**1.000**	**1.000**	0.136	0.110	17
PEKM $k = 8$	**0.996**	**0.995**	**0.992**	**1.000**	**1.000**	**1.000**	0.123	17
PEKM $k = 12$	1.000	1.000	1.000	1.000	1.000	1.000	0.124	17
PEKM-S $\alpha = 0.5$	0.878	0.854	0.756	0.811	0.982	0.948	0.153	31
PEKM-S $\alpha = 1$	0.897	0.873	0.770	0.814	1.000	1.000	0.265	63

Table S5: Segmentation F1 under illumination shifts (front and wrist cameras together; 6396 instances per shifted condition, 18,301 nominal). Colour thresholds as calibrated under nominal lighting. SAM backbone ViT-H.

Condition	HSV thresholds	SAM-FDM (colour prompts)	SAM-GTBOX F1	SAM-GTBOX IoU
Nominal	0.987	0.989	0.999	0.979
Direct sun (bright)	0.079	0.096	0.999	0.979
Overcast / dusk (dark)	0.000	0.000	0.999	0.979
Low sun (warm)	0.018	0.031	0.999	0.979

Closed-loop metrics. Success rate (primary) uses the task conditions of Section 3. *Harvest completion*: fruits correctly in the basket divided by the target number. *Wrong-pick rate*: episodes with an unripe fruit in the basket (FruitOccludedRipe). *Over-pick rate*: episodes with more than N fruits in the basket (FruitCount). *Drop rate*: episodes in which a fruit is dropped outside the basket. *Steps* to completion and steps per pick. *Engineering-error rate*: episodes terminated by infrastructure faults, reported separately and not counted as policy failures.

Event metrics. A detected event $[t_s, t^{\mathrm{r}}]$ is matched one-to-one, in time order, to a ground-truth detachment (the first frame at which the fruit is no longer attached) if the detachment falls within $[t_s - 3, t^{\mathrm{r}} + 3]$. We report precision, recall and F1; the count exact rate (fraction of episodes in which the number of detected events equals the number of picks); and the contact lead and release error in frames relative to detachment.

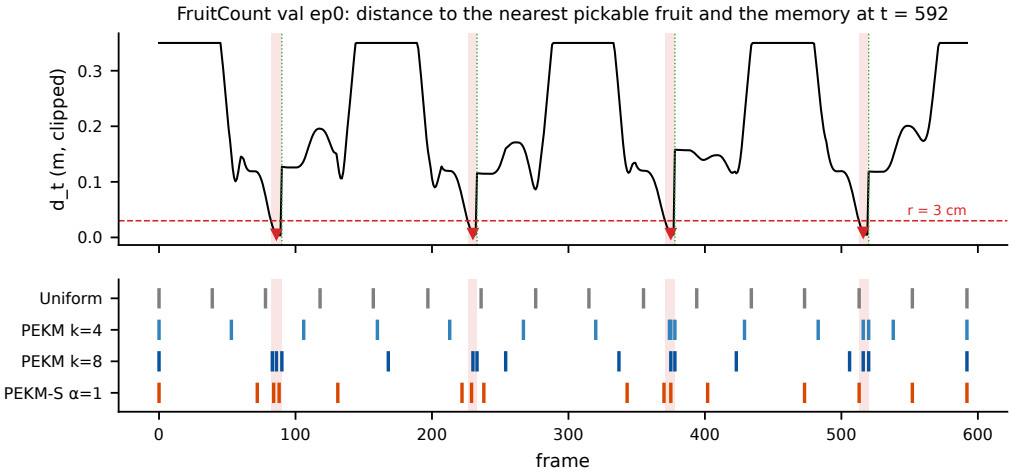


Figure S1: Memory timeline of one FruitCount validation episode. Top: the TCP-to-fruit distance d_t with detected events (shaded), contact keyframes (triangles), the radius r (dashed) and fruit detachments (dotted). Bottom: the 16 frames held by uniform memory, PEKM ($k = 4$, $k = 8$) and PEKM-S ($\alpha = 1$) at the last decision.

Table S6: Fruit anchors from SAM-FDM (ViT-H) on 199 validation episodes, matched to the true fruit positions at frame 0 (5 cm gate). Hidden ripe: fraction of ripe fruits later hidden by leaves that have an anchor ($n = 138$).

Anchors	Recall	Precision	Error med / p90 (cm)	Ripeness acc.	Hidden ripe
Frame 0 (before gust, ours)	0.998	0.995	0.48 / 0.58	1.000	**1.000**
Frame 0 + depth noise	0.998	0.995	0.45 / 0.56	1.000	1.000
Frame 29 (after gust)	0.783	0.993	0.49 / 0.58	1.000	0.000

Memory metrics. Memories are evaluated at every decision point (every 16 frames). A completed pick is covered if the memory holds a frame within ±3 frames of its contact or release keyframe. Event coverage: fraction of completed picks covered. Run coverage: at least one memory frame inside the pick's near run. Count recoverability: fraction of decisions at which every completed pick is covered. Recency coverage: coverage of the last, second and third most recent pick. Near-frame fraction and median gap between memory frames measure the cost in temporal coverage. For perturbed or perceived signals, the Jaccard similarity to the ground-truth memory and the recall of ground-truth keyframes within ±3 frames.

Table S7: Open-loop (teacher-forced) accuracy of a uniform-memory policy trained on clean demonstrations (10 episodes per set). TCP error at horizons 1, 8 and 16 of the action chunk; near: horizon 16 while $d_t \leq 10$ cm.

Set	Task	TCP error h1 / h8 / h16 (cm)	Near h16 (cm)	Gripper agreement
Train	FruitCount	0.44 / 1.23 / 1.83	0.75	99.5%
Train	FruitOccludedRipe	0.42 / 1.25 / 1.66	0.70	99.2%
Validation	FruitCount	0.51 / 1.37 / 2.38	1.14	98.0%
Validation	FruitOccludedRipe	0.40 / 1.32 / 1.75	0.78	98.9%

Table S8: Memory ablations of the base VLA on a general manipulation-memory benchmark (paired bootstrap, Holm-adjusted).

Intervention	Result
No history memory (pattern-recall task)	66.7% $\rightarrow$ 1.7% (-65.0 pp, 95% CI [-81.7, -46.7], $p < 0.001$)
No memory (video-swap task)	-11.7 pp (95% CI [-20.0, -3.3], $p = 0.006$)
Budget 32 $\rightarrow$ 16 frames (pattern recall)	64.4% $\rightarrow$ 26.7% (-37.8 pp, $p < 0.001$)
Single-slot replacement (counterfactual action loss)	Slot ranking reliability 0.97–0.99 over four tasks; cross-fitted relative gain 2.5–12.4%

Perception metrics. Instance precision, recall and F1 at IoU ≥ 0.5; mean IoU and ripeness accuracy of matched instances; recall stratified by the visible fraction of the fruit (>75%, 25–75%, <25%); latency per image on an L20 GPU; robustness to illumination (Section S5). Anchors: recall and precision against the true fruit positions at frame 0 (5 cm gate), median and 90th-percentile localization error, ripeness accuracy, and the fraction of leaf-hidden ripe fruits that have an anchor. Distances: error of $\hat{d}_t$ in the near range ($d_t \leq 10$ cm). Per-frame variant: the current frame's colour detections are back-projected, and an invisible fruit is unknown.

Closed-loop protocol details. Policy sampling seeds are 7, 8 and 9. All conditions of an episode and seed run in one block with the same environment seed and initial state, in randomized order, on policy servers whose identity is logged; each server holds a fixed group of conditions so that their checkpoints fit in GPU memory. The development gate requires a K0 success rate above 5% and at most 2 engineering-error episodes on the development set. Confidence intervals use a cluster bootstrap that resamples episodes and keeps all seeds and conditions of an episode together (20,000 resamples, fixed random seed).

Table S9: Provenance of all method parameters. No parameter was tuned on validation or test data.

Parameter	Value	Source
Memory budget B	16 frames	Fine-tuning configuration of the base model
Keyframe cap k	4, 8	Training-set event counts (at most 3 / 5 picks per episode)
Event radius r, tolerance tol	3 cm, 5 mm	Training-set sensitivity (E2); fruit radius 2.5–2.9 cm
Hysteresis exit, minimum run	$1.5r$, 3 frames	Fixed a priori
Coverage window	$\pm$3 frames	Fixed a priori (matching window of E2)
PEKM-S τ_d, recent slots	5 cm, 4	Fixed a priori
DART-H σ, τ, phase scales	0.04 rad, 10, (1, 0.5, 1, 1)	Calibration on training generation: TCP displacement $\approx$2.5 cm
HSV thresholds	see Section S5	Pixel F1 on 20 rendered training episodes
SAM prompt (box pad, point)	2 px, centroid	Instance F1 on rendered training episodes
Vacancy / occlusion margins	5 cm / 2 cm, 3 frames	Physical scale (fruit diameter $\approx$5 cm)
Grasp cue	2 frames, within r	Gripper geometry (empty-closed vs open)
Training (steps, batch, LR, EMA)	12k, 64, 2.5×10^{-5}, 0.999	Base-model fine-tuning recipe, unchanged

Training details. 12,000 steps; global batch size 64; AdamW with gradient clipping at 1.0; 500 warm-up steps to a peak learning rate of 2.5×10^{-5}, then cosine decay to 2.5×10^{-6}; EMA decay 0.999; frozen vision encoder; seed 42. Normalization statistics are recomputed on HarvestMem-Sim because the RM75-B joint ranges differ from those of the pretraining robot. Each condition is trained with fully sharded data parallelism on four NVIDIA L20 (48 GB) GPUs. DART-H acceptance rates were 0.57 (FruitCount) and 0.66 (FruitOccludedRipe).

Implementation. The simulation uses ManiSkill3 (Tao et al., 2025) with SAPIEN rendering and mplib (Hao Su Lab, 2024) motion planning. The policy is implemented in JAX on the openpi code base (Black et al., 2024), and SAM in PyTorch with the official checkpoints. Experiments ran on a server with eight NVIDIA L20 (48 GB) GPUs; in closed-loop evaluation,

policy servers run on GPUs separate from the simulators.

References

Black, K., Brown, N., Driess, D., Esmail, A., Equi, M., Finn, C., Fusai, N., Groom, L., Hausman, K., Ichter, B., et al., 2024. π_0: A vision-language-action flow model for general robot control. arXiv preprint arXiv:2410.24164 .

Hao Su Lab, 2024. MPlib: A lightweight motion planning library. https://github.com/haosulab/MPlib. Accessed 2026-10-28.

Laskey, M., Lee, J., Fox, R., Dragan, A., Goldberg, K., 2017. DART: Noise injection for robust imitation learning, in: Proceedings of the Conference on Robot Learning (CoRL), pp. 143–156.

Tao, S., Xiang, F., Shukla, A., Qin, Y., Hinrichsen, X., Yuan, X., Bao, C., Lin, X., Liu, Y., Chan, T.k., et al., 2025. ManiSkill3: GPU parallelized robotics simulation and rendering for generalizable embodied AI, in: Proceedings of Robotics: Science and Systems (RSS).